GDG on Campus
GRIET

Category: Web Development & Social Impact

Problem Statement 1:
LawLink - Gamified Legal Literacy Platform

Description

Legal information is often complex and scattered, leaving students and young adults unaware of their rights and responsibilities. Build an interactive platform that simplifies essential Indian laws through real-life scenarios, short explanations, quizzes, and challenges. For each issue, the platform should explain applicable rights, practical next steps, and relevant authorities or helplines.

Requirements

- Develop a responsive web or mobile application with user login and progress tracking.
- Cover topics such as cybercrime, consumer rights, road laws, digital privacy, fundamental rights, and student, workplace, and women's safety.
- Include scenario-based learning, quizzes, points, badges, levels, and a leaderboard.
- Provide search or chatbot guidance plus a verified emergency-contact and legal-resource directory.
- Clearly state that the platform offers legal awareness only and is not a substitute for professional legal advice.

Expected Outcomes

- A functional prototype that makes legal education practical, accessible, and engaging.
- A clear demonstration of core modules, gamification, trusted resources, and real-world usability.
- Bonus: regional languages, voice queries, personalized recommendations, location-based legal aid, or verified AI-generated scenarios.

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Category: AI & Machine Learning

Problem Statement 1:
"VeriTrust AI" – Dual-Agent "Maker & Judge" Hallucination Guardrail System

Description

Develop an AI-based software solution that acts as a real-time compliance filter to detect and eliminate AI hallucinations in automated customer service workflows. The system leverages a dual-agent graph architecture where a "Maker Agent" generates conversational drafts from company manuals, and a strict "Judge Agent" intercepts and cross-checks the text against a verified corporate database before the message is displayed to a client. The system should provide real-time metrics on accuracy drifts via a unified administrative dashboard.

Requirements

- Agentic workflows and graphs (LangGraph or CrewAI)
- Large Language Models (OpenAI API, Anthropic API, or local Ollama)
- Database retrieval (LlamaIndex, ChromaDB, or SQLite)
- Web dashboard development (React, FastAPI or Streamlit)

Expected Outcomes

- A working prototype containing a multi-agent loop that captures, evaluates, and automatically corrects inaccurate or hallucinated AI outputs.
- A clear demo showcasing factual correction speed, system latency, and a side-by-side dashboard interface tracking blocked vs. approved responses, supported by accuracy metrics.

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Category: AI/ML & Data Science

Problem Statement 1:
Campus Lost and Found Intelligence

Description

Every day, students lose personal belongings such as ID cards, laptops, books, gadgets, or accessories on campus, and reuniting them with their owners is often slow and manual. Develop an AI-powered platform that helps in matching lost items with found items by analyzing images and textual descriptions. The system should automatically identify item categories, extract relevant attributes (e.g., brand, color, shape, distinguishing features) and suggest the most likely matches with a confidence score. It should provide a simple and intuitive interface for users to report lost or found items, search for potential matches, and facilitate the recovery process.

Requirements

- Use computer vision and/or deep learning for image analysis.
- Use NLP for processing textual descriptions.
- Develop a web or mobile interface (any tech stack).
- Participants may use open datasets or create their own dataset.
- Additional features such as location and time information, similarity ranking, notifications, and a claim verification mechanism are encouraged.

Expected Outcomes

- A working prototype that allows users to report lost and found items, automatically finds potential matches with confidence scores, and provides an intuitive user experience.
- A clear demonstration of the AI/ML approach, key features, and real-world applicability, supported by relevant evaluation metrics (e.g., matching accuracy, precision@k).
- Bonus points for creative features such as item tracking history, community insights, or automated verification mechanisms.

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Category: AI Security & Infrastructure

Problem Statement 1: TrustMesh — Dynamic Agent-to-Agent Authorization Protocols

Problem Statement

Design and build a secure, standard protocol that allows AI agents to prove their identity and authority to each other at runtime. Create dynamic access controls to prevent unauthorized data access or actions between interacting autonomous agents. Develop an open-source framework or protocol that enables autonomous AI agents, operating without direct human control, to establish mutual trust. As agents increase in autonomy (e.g., booking flights, accessing APIs), they cannot assume other agents are legitimate. This system should allow agents from different frameworks (e.g., CrewAI, AutoGen) to securely verify each other's credentials and permissions before sharing information or executing delegated tasks, without needing human intervention for each transaction.

Requirements

- Cryptographic key management for agent identity (e.g., DIDs, signed tokens), a definition language for agent capability scopes (e.g., spend limits, data access), interoperability with major agent frameworks, and robust mechanisms to prevent replay attacks, privilege escalation, and payload spoofing.

Expected Outcomes

- A working protocol or middleware that demonstrates a secure verification handshake between at least two agents.
- A dynamic policy enforcement engine that visually blocks unauthorized or out-of-scope agent requests.
- A security validation harness with simulated attack scenarios (e.g., impersonation, token reuse).
- Formal protocol documentation with message schemas and sequence diagrams.

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Category: Open Innovation

Problem Statement:
Open Innovation — Build Your Own Solution

Description

Bring your own problem, idea, or prototype and turn it into a working solution. Open Innovation gives participants the freedom to identify a real-world problem, define the opportunity, and build a technology-driven solution of their choice. Teams may start from a problem statement, an early-stage concept, or an existing prototype and are encouraged to improve, validate, and demonstrate its real-world impact.

Requirements

- Participant-defined problem statement or clearly identified real-world problem
- A proposed solution with a clear use case, target users, and expected impact
- Working prototype, MVP, or meaningful implementation demonstrating the idea
- Use any appropriate technology, framework, platform, or AI/ML approach

Expected Outcomes

- A functional prototype or MVP that addresses the team's chosen problem and demonstrates a clear solution workflow.
- A compelling demo explaining the problem, solution, innovation, technical implementation, and potential real-world impact.